

2015-2016 学年第一学期期末考试试卷

一、解以下一阶微分方程. (40 分)

1. $\frac{dy}{dx} = \frac{x}{x^2 - 2y + 1}$ (10 分)

2. $(\cos x - x \sin x)y dx + (x \cos x - 2y)dy = 0$ (10 分)

3. $2dx + \sqrt{\frac{x}{y}}dy - \sqrt{\frac{y}{x}}dx = 0$ (10 分)

4. $2x\frac{dy}{dx} + y = y^2\sqrt{x - x^2y^2}$ (10 分)

二、解以下常系数微分方程. (20 分)

1. $\frac{d^2 y}{dx^2} - 4\frac{dy}{dx} + 5y = e^{2x} \cos x + \sin x$ (10 分)

2. $\frac{d^3 y}{dx^3} - \frac{d^2 y}{dx^2} - \frac{dy}{dx} + y = xe^x$ (10 分)

三、解以下变系数微分方程. (20 分)

1. $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = \frac{x^2}{y} \left(\frac{dy}{dx} \right)^2$ (10 分)

2. $(x-1) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + y = (x-1)^2$ (10 分)

四、解以下一阶线性微分方程组. (20 分)

$$1. \begin{cases} 2\frac{dx}{dt} + \frac{dy}{dt} = x + y + e^t \\ \frac{dx}{dt} + 2\frac{dy}{dt} = -x + y \end{cases}$$

$$2. \begin{cases} \frac{dx}{dt} = 2x - y + 2z \\ \frac{dy}{dt} = x + 2z \\ \frac{dz}{dt} = y - 2x - z \end{cases}$$

2015-2016 学年第一学期期末考试试卷参考答案

一、解以下一阶微分方程.

1、【解析】方程可化为 $(x^2 - 2y + 1)dy - xdx = 0$. 记 $N = x^2 - 2y + 1$, $M = -x$

$$\mu(y) = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{-M} = -2, \text{ 关于 } y \text{ 的积分因子 } G(y) = e^{\int \mu(y) dy} = e^{-2y}$$

$$\text{则 } (x^2 - 2y + 1)e^{-2y} dy - xe^{-2y} dx = 0 \Rightarrow -\frac{x^2}{2}e^{-2y} + ye^{-2y} = C$$

$$\text{故原方程的解为 } y - \frac{x^2}{2} = Ce^{2y}.$$

【考点延伸】积分因子法求解微分方程.

2、【解析】由于 $(\cos x - x \sin x)ydx + (x \cos x - 2y)dy = d(x \cos x \cdot y - y^2)$

$$\text{故原方程的解为 } xy \cos x - y^2 = C.$$

【考点延伸】凑微分法求解微分方程.

3、【解析】当 $x > 0$ 时, 方程即为 $\left(2 - \sqrt{\frac{y}{x}}\right)dx + \sqrt{\frac{x}{y}}dy = 0$. 令 $M = 2 - \sqrt{\frac{y}{x}}$, $N = \sqrt{\frac{x}{y}}$

$$\text{则 } \mu(x) = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{-\frac{1}{\sqrt{xy}}}{\sqrt{\frac{x}{y}}} = -\frac{1}{x}, \text{ 关于 } x \text{ 的积分因子 } G(x) = e^{\int \mu(x) dx} = \frac{1}{x}.$$

$$\text{则有 } \left(\frac{2}{x} - \frac{1}{x}\sqrt{\frac{y}{x}}\right)dx + \frac{1}{x}\sqrt{\frac{x}{y}}dy = 0 \Rightarrow 2\ln x + 2\sqrt{\frac{y}{x}} = C$$

$$\text{当 } x < 0 \text{ 时, 令 } x' = -x, y' = -y, \text{ 方程化为 } \left(2 - \sqrt{\frac{y'}{x'}}\right)dx' + \sqrt{\frac{x'}{y'}}dy' = 0.$$

$$\text{解得 } 2\ln x' + 2\sqrt{\frac{y'}{x'}} = C \Rightarrow 2\ln(-x) + 2\sqrt{\frac{y}{x}} = C$$

易知 $x \neq 0$

$$\text{故原方程的解为 } y = x(\ln|x| + C)^2.$$

【考点延伸】积分因子法求解微分方程.

4、【解析】令 $xy^2 = u$, 则 $\frac{du}{dx} = y^2 + 2xy\frac{dy}{dx}$

当 $x = 0$ 时, $y = 0$.

当 $y = 0$ 时, 满足方程.

当 $x \neq 0, y \neq 0$ 时, 方程可化为 $\frac{\frac{du}{dx} - y^2}{2xy} + \frac{y}{2x} = \frac{y^2}{2x} \sqrt{x - x^2 y^2}$,

$$\text{即 } \frac{du}{dx} = y^3 \sqrt{x - x^2 y^2} = \begin{cases} \frac{u}{x} \sqrt{u - u^2}, & y > 0 \\ -\frac{u}{x} \sqrt{u - u^2}, & y < 0 \end{cases}$$

$$\text{分离变量: } \int \frac{1}{u \sqrt{u - u^2}} du = \begin{cases} \int \frac{1}{x} dx, & y > 0 \\ -\int \frac{1}{x} dx, & y < 0 \end{cases} \Rightarrow -\frac{2\sqrt{u - u^2}}{u} = \begin{cases} \ln x + C, & y > 0 \\ -\ln x + C, & y < 0 \end{cases}$$

$$\text{从而方程的解为 } -\frac{2\sqrt{xy^2 - x^2 y^4}}{xy^2} = \begin{cases} \ln x + C, & y > 0 \\ -\ln x + C, & y < 0 \end{cases}.$$

$$\text{综合起来可得方程解为: } -\frac{2y\sqrt{x - x^2 y^2}}{xy^2} = \ln x + C.$$

【考点延伸】可分离变量微分方程.

二、解以下常系数微分方程.

1、【解析】由特征方程 $\lambda^2 - 4\lambda + 5 = 0$, 得 $\lambda_{1,2} = 2 \pm i$

故齐次方程的通解为 $Y = e^{2x}(C_1 \cos x + C_2 \sin x)$

设 $y^* = xe^{2x}(A \cos x + B \sin x) + (C \cos x + D \sin x)$, 代入方程可得

$$2e^{2x}(B \cos x - A \sin x) + 4(C - D) \cos x + 4(C + D) \sin x = e^{2x} \cos x + \sin x$$

$$\text{故有 } A = 0, B = \frac{1}{2}, C = \frac{1}{8}, D = \frac{1}{8}, \text{ 故 } y^* = \frac{1}{2}xe^{2x} \sin x + \frac{1}{8}(\cos x + \sin x)$$

$$\text{因此原方程的通解为 } y = e^{2x}(C_1 \cos x + C_2 \sin x) + \frac{1}{2}xe^{2x} \sin x + \frac{1}{8}(\cos x + \sin x).$$

【考点延伸】二阶非齐次线性微分方程.

2、【解析】由特征方程 $\lambda^3 - \lambda^2 - \lambda + 1 = 0$, 得 $\lambda_1 = \lambda_2 = 1, \lambda_3 = -1$

故齐次方程的通解为 $Y = C_1 e^{-x} + (C_2 + C_3 x)e^x$

设 $y^* = x^2(Ax + B)e^x$, 代入方程可得 $12Ax + 6A + 4B = x$

$$\text{故有 } A = \frac{1}{12}, B = -\frac{1}{8} \Rightarrow y^* = x^2 \left(\frac{1}{12}x - \frac{1}{8} \right) e^x$$

$$\text{因此原方程的通解为 } y = C_1 e^{-x} + (C_2 + C_3 x)e^x + x^2 \left(\frac{1}{12}x - \frac{1}{8} \right) e^x.$$

【考点延伸】高阶非齐次线性微分方程.

三、解以下变系数微分方程.

1、【解析】令 $x = e^t$, $t = \ln x$, 则有 $\frac{dy}{dx} = \frac{1}{x} \frac{dy}{dt}$, $\frac{d^2y}{dx^2} = \frac{1}{x^2} \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right)$

$$\text{方程可化为 } \frac{d^2y}{dt^2} - \frac{1}{y} \left(\frac{dy}{dt} \right)^2 + y = 0.$$

$$\text{再令 } p = \frac{dy}{dt}, \text{ 则 } \frac{d^2y}{dt^2} = \frac{dp}{dy} \frac{dy}{dt} = p \frac{dp}{dy}$$

$$\text{方程可化为 } \frac{dp}{dy} - \frac{p}{y} + \frac{y}{p} = 0$$

$$\text{最后令 } q = \frac{p}{y}, \text{ 则方程化为 } y \frac{dq}{dy} = -\frac{1}{q} \Rightarrow q = \pm \sqrt{-2\ln|y| + C_1}$$

$$\Rightarrow p = \pm y \sqrt{-2\ln|y| + C_1} \Rightarrow \pm \frac{dy}{y \sqrt{-2\ln|y| + C_1}} = dt$$

$$\text{得 } -\sqrt{-2\ln|y| + C_1} = t + C_2 \text{ 或 } \sqrt{-2\ln|y| + C_1} = t + C_2$$

$$\Rightarrow |y| = e^{-\frac{(t+C_2)^2 - C_1}{2}} = e^{-\frac{(\ln x + C_2)^2 - C_1}{2}}$$

【考点延伸】欧拉方程.

2、【解析】方程化为 $\frac{d^2y}{dx^2} - \frac{x}{x-1} \frac{dy}{dx} + \frac{1}{x-1} y = x-1$

显然 $y_1^* = -1 - x^2$ 为一特解, $y_1 = x$ 为一齐次解.

$$\text{则 } y_2 = \int \frac{1}{y_1^2} e^{-\int P(x) dx} dx = \int \frac{x-1}{x^2} e^x dx = \frac{e^x}{x}.$$

故原方程的解为 $y = y_1(C_1 + C_2 y_2) + y_1^* = C_1 x + C_2 e^x - x^2 - 1$.

【考点延伸】刘维尔公式求解二阶变系数微分方程.

四、解以下一阶线性微分方程组.

$$1、【解析】 \text{记 } \begin{cases} 2 \frac{dx}{dt} + \frac{dy}{dt} = x + y + e^t & (1) \\ \frac{dx}{dt} + 2 \frac{dy}{dt} = -x + y & (2) \end{cases}$$

$$2 \times (1) - (2) \text{ 得 } 3 \frac{dx}{dt} = 3x + y + 2e^t \Rightarrow y = 3 \frac{dx}{dt} - 3x - 2e^t$$

$$\text{代入(1)中得 } 2 \frac{dx}{dt} + 3 \frac{d^2x}{dt^2} - 3 \frac{dx}{dt} - 2e^t = x + 3 \frac{dx}{dt} - 3x - 2e^t + e^t$$

$$\text{整理得 } 3 \frac{d^2x}{dt^2} - 4 \frac{dx}{dt} + 2x = e^t$$

由特征方程 $3\lambda^2 - 4\lambda + 2 = 0 \Rightarrow \lambda_{1,2} = \frac{2}{3} \pm \frac{\sqrt{2}}{3}i$

故齐次方程的通解为 $X = e^{\frac{2}{3}t} \left(C_1 \cos \frac{\sqrt{2}}{3}t + C_2 \sin \frac{\sqrt{2}}{3}t \right)$

设特解 $x^* = Ae^t$, 代入方程解得 $A=1 \Rightarrow x^* = e^t$.

故 $x = X + x^* = e^{\frac{2}{3}t} \left(C_1 \cos \frac{\sqrt{2}}{3}t + C_2 \sin \frac{\sqrt{2}}{3}t \right) + e^t$.

从而 $y = e^{\frac{2}{3}t} \left[(-C_1 + \sqrt{2}C_2) \cos \frac{\sqrt{2}}{3}t - (C_2 + \sqrt{2}C_1) \sin \frac{\sqrt{2}}{3}t \right] - 2e^t$.

【考点延伸】常微分方程组.

2、【解析】令 $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, $A = \begin{pmatrix} 2 & -1 & 2 \\ 1 & 0 & 2 \\ -2 & 1 & -1 \end{pmatrix}$, 则 $\frac{dX}{dt} = AX$.

$$|A - \lambda E| = \begin{vmatrix} 2-\lambda & -1 & 2 \\ 1 & -\lambda & 2 \\ -2 & 1 & -1-\lambda \end{vmatrix} = -(\lambda^2 + 1)(\lambda - 2) = 0 \Rightarrow \lambda_1 = 2, \lambda_{2,3} = \pm i.$$

$$\text{故 } X_1 = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} e^{2t}, X_2 = \begin{pmatrix} 1+i \\ 1+i \\ -1 \end{pmatrix} e^{it}, X_3 = \begin{pmatrix} 1-i \\ 1-i \\ -1 \end{pmatrix} e^{-it}$$

原方程组的解为 $X = C_1 X_1 + C_2 X_2 + C_3 X_3$.

【考点延伸】常微分方程组.